ORLA 2.0: Modernizing an App-Based Aphasia Rehabilitation Tool

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Figure 1: Comparison of the ORLA program currently in distribution (left) to the new ORLA beta (right).

ORLA (Oral Reading for Language in Aphasia):

systematic rehabilitation program that targets reading comprehension and other language modalities. (Cherney et al., 1986)

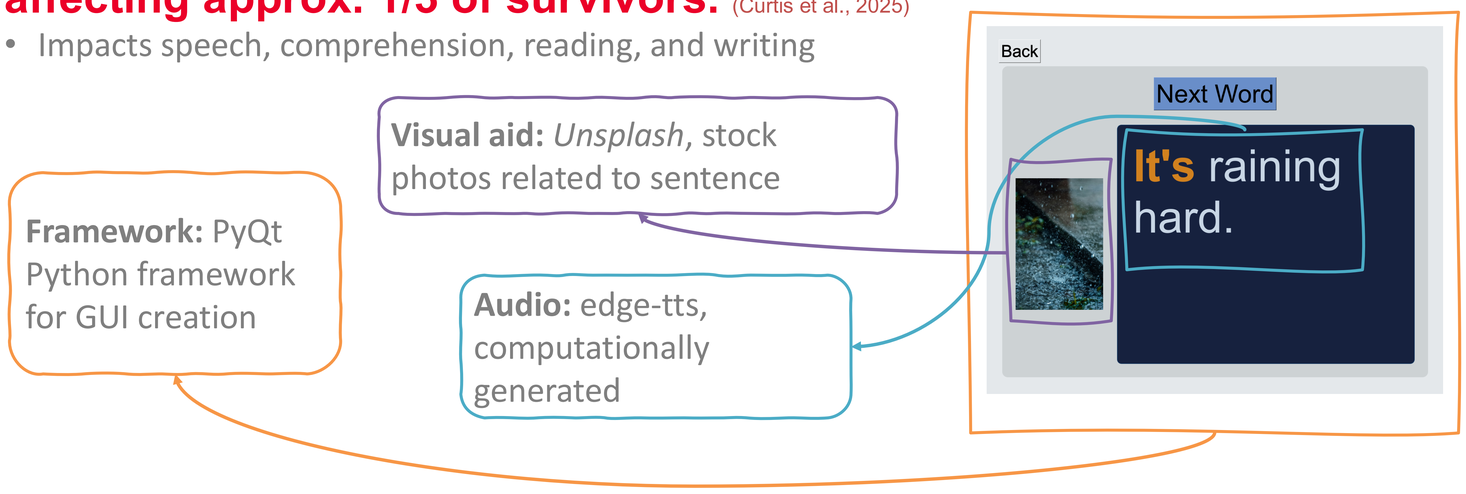
During an ORLA session, the person with aphasia repeatedly reads sentences or paragraphs aloud following these steps:

- **Look and Listen:** Clinician reads the text aloud while pointing to each word.
- **Joint Pointing:** Clinician reads aloud again, patient points
- **Choral Reading:** Clinician and patient read and point together
- **Word Identification:** Clinician states a word in the sentence, patient points
- **Word Production:** Clinician points to a word, patient reads aloud independently.
- **Final Choral Reading:** Patient and clinician read the entire text aloud in unison.

The program is also graded into four levels of difficulty based on text length and reading level, allowing the clinician to adjust for a range of aphasia severities. (Cherney, 2010)

Aphasia: acquired language impairment, most caused by stroke, affecting approx. 1/3 of survivors. (Curtis et al., 2025)

- Impacts speech, comprehension, reading, and writing



Future Steps

Development is currently in beta. There are multiple future directions for this project:

- **Update text-to-speech:** edge-tts is robotic, update AI model for natural speech
- **Allow personalization:** Give the user the ability to choose the voices that read sentences, change theme, etc.
- **Gamification:** Increase motivation by adding features that ”gamify” the experience using a point system, intermittent reward, etc.

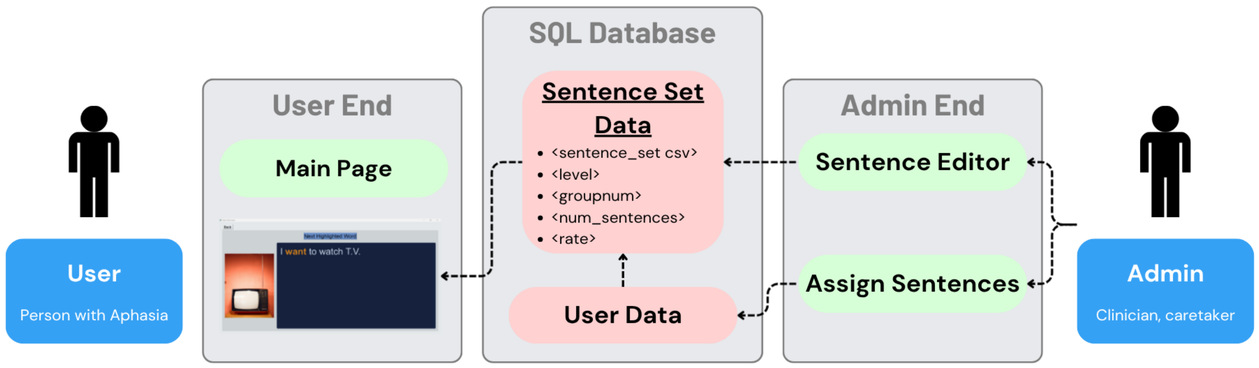


Figure 2. Data flow diagram showing the interface between admin and user. The admin creates sentence sets and assigns them to users, this data is stored in the backend database, and the user only sees the sentences without access to any of the data.

References

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3. Curtis, H., Bircanin, F., & Neate, T. (2025). "An old bastard in bright orange satin!": Zuzenna's aphasia diary: And lessons learned from DIY augmentative and alternative communication. *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '25) (Article 48, pp. 1–20). Association for Computing Machinery. https://doi.org/10.1145/3663547.3746369

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